>_ Nixpert

Online training in Linux, Infrastructure, and Applied AI

Infrastructure Course Track

The physical and virtual stack behind the modern data center.

Live cohorts · Real labs · Cert-aligned · Nixpert certificate

```
$ nixpert enroll --track infrastructure
✓ Hardware Basics ................ ready
✓ Hyperconverged (HCI) ........... ready
✓ Cloud .......................... ready
$ _
```

nixpert.in · hello@nixpert.in · Live, hands-on, practitioner-led

Course Catalog · 2026 Cohorts

Track Overview

Three courses covering the physical and virtual stack that runs the modern data center: hardware, cloud, and hyperconverged infrastructure. Take them individually or as the Infrastructure Career Bundle.

	Hardware Basics	Hyperconverged (HCI) Basics	Cloud
Duration	5 weeks	5 weeks	6 weeks
Live sessions	10	10	12
Level	Beginner	Beginner to Intermediate	Beginner to Pro
Prerequisites	None	Basic server and virtualization ideas	None
Cert alignment	CompTIA A+ / Hardware+	NCP / VCP foundations	Cloud provider fundamentals
Capstone	Diagnose and rebuild a faulty server	Design a 3-node HCI cluster	Deploy a 3-tier app with IaC
Launch price	**₹ 2,999**	**₹ 4,999**	**₹ 6,499**
List price	**₹ 3,999**	**₹ 5,999**	**₹ 7,999**

Every cohort includes

- Weekly live classes with Q&A, recorded and available for 12 months.
- Hands-on labs for every module, with 1:1 support when a lab fights back.
- Private community chat and weekly office hours.
- A graded portfolio capstone and a Nixpert completion certificate.
- Upgrade credit: alumni move to a bigger course or bundle by paying only the difference plus ₹ 500.

Hardware Basics

5 weeks · 10 live sessions · Beginner · ₹ 3,999 (launch ₹ 2,999)

CPUs, memory, storage, and the rest of the metal, aligned with CompTIA A+ and Hardware+ objectives. You leave able to spec, diagnose, and maintain desktop and server hardware.

Audience	Format	Outcome
Freshers, desktop support, NOC and DC technicians	2 live sessions / week + teardown demos	Spec, diagnose, and maintain real hardware confidently

Week 1 · CPU and Memory

- CPU architecture in practice: cores, threads, cache, sockets, and clock speeds.
- Cooling: air, liquid, thermal paste, and thermal throttling.
- RAM: DDR generations, channels, ECC, and capacity planning.

Lab: Read real spec sheets and match the right CPU and RAM to three workload briefs.

Week 2 · Motherboards, Firmware, and Power

- Motherboard anatomy: chipsets, PCIe lanes, slots, and headers.
- BIOS and UEFI: boot order, secure boot, and firmware updates.
- Power supplies: wattage, rails, efficiency ratings, and redundancy.

Lab: Configure UEFI settings for a server build and validate POST behaviour.

Week 3 · Storage Hardware

- HDD vs SSD vs NVMe: interfaces, form factors, and endurance.
- RAID levels in hardware: controllers, cache, and battery backup.
- Drive health: SMART data and predicting failure.

Lab: Build a RAID array, pull a disk mid-write, and walk the rebuild.

Week 4 · Servers and Networking Hardware

- Server form factors: tower, rack, and blade. Rails, U sizing, and cabling discipline.
- Out-of-band management: BMC, IPMI, iDRAC, and iLO.
- NICs, switches, and transceivers: speeds, ports, and link issues.

Lab: Bring up a server entirely through its BMC, no monitor attached.

Week 5 · Diagnostics and Maintenance

- A structured hardware troubleshooting method: symptoms to component.
- POST codes, beep codes, memtest, and burn-in testing.
- Preventive maintenance, ESD safety, and spare-part strategy.

Capstone

Diagnose a deliberately faulty server, identify every bad component with evidence, and document the rebuild as a professional service report.

Cloud

6 weeks · 12 live sessions · Beginner to Pro · ₹ 7,999 (launch ₹ 6,499)

Core cloud operations across the major providers: compute, storage, identity, networking, and the cost discipline that keeps the CFO calm. Concepts are taught provider-agnostic, labs run on real accounts.

Audience	Format	Outcome
Sysadmins, support engineers, and cloud career switchers	2 live sessions / week + live cloud labs	Operate workloads across AWS, Azure, and GCP

Week 1 · Cloud Foundations and Identity

- Service models and shared responsibility: what you manage vs what the provider manages.
- Regions, availability zones, and designing for failure.
- IAM: users, roles, policies, and least privilege across providers.

Lab: Set up a secure account baseline: MFA, roles, budgets, and alerts.

Week 2 · Compute

- VM families and sizing, images, and instance lifecycle.
- Load balancing and autoscaling: when and how.
- Spot and reserved capacity: paying less on purpose.

Lab: Deploy an autoscaling web tier behind a load balancer and watch it scale under load.

Week 3 · Storage

- Object, block, and file storage: choosing correctly.
- Snapshots, replication, and lifecycle policies.
- Durability vs availability, and what those nines really mean.

Lab: Build a tiered storage setup with lifecycle rules and restore from snapshot.

Week 4 · Networking

- VPCs, subnets, route tables, and gateways.
- Security groups and network ACLs: layered traffic control.
- DNS, private endpoints, and hybrid connectivity (VPN, peering).

Lab: Build a two-tier VPC with public web and private database subnets.

Week 5 · Operations and Cost

- Monitoring and logging: metrics, alarms, and dashboards.
- Cost management: tagging, budgets, rightsizing, and the bill no one expected.
- Backups and disaster recovery patterns.

Lab: Audit a deliberately wasteful lab account and cut its monthly cost by half.

Week 6 · Infrastructure as Code and Capstone

- IaC fundamentals with Terraform: providers, state, plan, and apply.
- From console clicks to repeatable code.

Capstone

Deploy a three-tier application (load balancer, app tier, database) with Terraform, complete with monitoring, backups, and a cost estimate, then tear it down cleanly.

Hyperconverged (HCI) Basics

5 weeks · 10 live sessions · Beginner to Intermediate · ₹ 5,999 (launch ₹ 4,999)

Vendor-agnostic hyperconverged fundamentals: how converged compute and storage nodes, distributed storage, and VM high availability actually work. The foundation for NCP and VCP certification tracks, taught by engineers who run HCI in production.

Audience	Prerequisite	Outcome
VMware/storage admins, DC engineers, presales	Basic server and virtualization concepts	Design and operate HCI clusters with confidence

Week 1 · From Three-Tier to HCI

- Why HCI exists: the pain of SANs, silos, and forklift upgrades.
- Anatomy of an HCI node: compute, local storage, and the controller layer.
- Hypervisor choices and where they fit: ESXi, AHV, Hyper-V, and KVM.

Lab: Map a legacy three-tier environment onto an equivalent HCI design.

Week 2 · Distributed Storage

- Clustered filesystems and distributed storage fabrics.
- Replication factors, data locality, and metadata.
- Deduplication, compression, and erasure coding trade-offs.

Lab: Trace a write through a simulated cluster: locality, replicas, and acknowledgement.

Week 3 · High Availability and Failure Handling

- VM HA: restart policies, admission control, and failure domains.
- What happens when a disk, node, or block fails: rebuilds and self-healing.
- Maintenance mode and rolling upgrades without downtime.

Lab: Fail a node in the lab cluster and narrate the recovery, step by step.

Week 4 · Operating an HCI Cluster

- Scaling out: adding nodes, storage-only nodes, and cluster expansion limits.
- Capacity planning: CPU, RAM, and storage runway.
- Monitoring, alerting, and performance baselining.

Lab: Produce a capacity plan and expansion proposal for a growing lab cluster.

Week 5 · Vendor Landscape and Capstone

- The HCI market mapped: Nutanix, VMware vSAN, Azure Stack HCI, and open-source options.
- How this course maps onto NCP and VCP certification tracks.

Capstone

Design a three-node HCI cluster for a given business brief: sizing, replication factor, failure domains, and growth plan, presented as an architecture review.

Pricing and Enrollment

Transparent pricing, launch discounts for the first cohort, and an upgrade path that rewards staying with Nixpert.

Course	Duration	List price	Launch price (first cohort)
Hardware Basics	5 weeks	₹ 3,999	**₹ 2,999**
Hyperconverged (HCI) Basics	5 weeks	₹ 5,999	**₹ 4,999**
Cloud	6 weeks	₹ 7,999	**₹ 6,499**
Infrastructure Career Bundle (all three courses)	**Flexible**	**₹ 14,999**	**₹ 11,999**

Offers and policies

- **Upgrade credit:** alumni of any course can move up by paying only the price difference plus ₹ 500.
- **Cert alignment:** Hardware Basics maps to CompTIA A+/Hardware+ and HCI Basics builds the foundation for NCP and VCP tracks.
- **Corporate and team training:** custom batches for teams of 5 or more. Write to hello@nixpert.in for a quote.
- **Payment:** UPI, cards, and net banking. EMI available on the flagship course and the bundle.
- **Refunds:** full refund up to 48 hours after the first live session, no questions asked.
- **Access:** recordings, labs, and community access for 12 months from enrollment.

How to enroll

- Visit **nixpert.in** and submit the enrollment form with your chosen course.
- We reply with the next cohort dates and an invite to a free intro call.
- Reserve your seat. Cohorts are capped to keep labs and 1:1 support meaningful.

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